Edge Computing Framework For Power Factor and Load Demand Management

Venkataramanan K, Professor,
Dept. Of Electrical and Electronics,
Prathyusha Engineering College, Thiruvallur,
raceramana@gmail.com

Janaki Raman K, UG Scholar,
Dept. Of Electrical and Electronics,
Prathyusha Engineering College, Thiruvallur,
janakiraman3679@gmail.com

Purushothaman M, UG Scholar,
Dept. Of Electrical and Electronics,
Prathyusha Engineering College, Thiruvallur,
mpurushothaman190@gmail.com

Raja Pandi P, UG Scholar,
Dept. Of Electrical and Electronics,
Prathyusha Engineering College, Thiruvallur,
rpandi84870@gmail.com

Abstract - The integration of highly variable industrial loads into modern microgrid environments necessitates rapid power quality regulation and intelligent energy distribution. Traditional centralized energy management systems often suffer from significant network propagation delays, which can hinder effective reactive power compensation during transient events. This paper presents the design and hardware implementation of an edge-computing control framework utilizing an ESP32 microcontroller for simultaneous Power Factor Correction (PFC) and Load Demand Management (LDM). The system architecture is designed to manage high-speed sensing and actuation tasks by interfacing a lightweight neural network algorithm. This setup enables the localized control of a multi-stage capacitor bank, achieving reactive power compensation within sub-cycle timeframes (<10 ms). Concurrently, the controller executes an LDM protocol that dynamically routes auxiliary renewable DC sources simulating solar and wind inputs based on predefined active power thresholds. System metrics are synchronized via MQTT telemetry to a remote Firebase Realtime Database for monitoring. The proposed control strategy was validated through MATLAB/Simulink simulations and verified on a physical hardware prototype. Experimental results demonstrate that the ESP32-based framework effectively manages load fluctuations, maintains a near-unity power factor, and significantly reduces actuation latency compared to cloud-dependent configurations, ensuring stable and autonomous microgrid operation.

Index Terms: Edge Computing, Power Factor Correction (PFC), Load Demand Management (LDM), ESP32, FreeRTOS, Neural Network, Microgrid.

I. Introduction

Power quality and grid stability are facing serious issues as nonlinear loads and intermittent renewable energy sources become more prevalent in localized microgrids. Motors and other inductive loads cause displacement power factor problems in industrial settings, increasing apparent power demand and perhaps resulting in utility fines **[Silva et al., 2025].** These problems have traditionally been resolved by centralized energy management systems and Automatic Power Factor Correction (APFC) panels. Nevertheless, these systems frequently depend on cloud-based controllers that experience severe network propagation delays or electromechanical contactors with high switching latency **[Kumar et al., 2025].**

An ESP32-based edge computing framework for real-time power quality control is proposed in this paper. The method achieves deterministic control over reactive power injection by moving the computational load from the cloud to the network edge **[Sharma et al., 2024]**. According to recent research by **[Wright et al. (2024)]** and **[Garcia et al. 2024]**, the architecture incorporates a neural network-based inference engine to forecast load transients and carry out multi-level capacitor switching in sub-cycle timeframes. Furthermore, by dynamically integrating auxiliary renewable sources, the system integrates an autonomous Load Demand Management (LDM) protocol to balance active power consumption **[Morozov et al., 2024].**

In order to address the latency constraints associated with cloud-dependent designs, this research proposes a localized, intelligent control strategy utilizing an Advanced Edge Computing framework based on the ESP32 architecture **[Nguyen et al., 2025; Rossi et al., 2025]**. The system accomplishes three main goals by carefully separating crucial, time-sensitive tasks from asynchronous network processes via parallelized task scheduling.

Sub-cycle Actuation: By implementing Neural Network-based Power Factor Correction (PFC) locally on the controller, it allows for almost immediate response. This satisfies the quick transient response requirements noted by Kumar et al. (2025) by guaranteeing that actuation decisions for the capacitor bank are computed and carried out in less than 10 ms. **Autonomous Resource Routing:** It automatically routes supplemental renewable DC sources by assessing real-time active power consumption against critical thresholds (20W and 40W). This is consistent with reactive power management techniques for smart grids that are optimized using deep learning [Khan et al., 2025]. **Non-Blocking Cloud Telemetry:** Using the MQTT protocol and a Firebase Realtime Database, it offers constant system visibility. The architecture makes guarantee that high-speed local control loops are not slowed down or interfered with by high-latency cloud connection [Sharma et al., 2024]. Ultimately, this approach guarantees immediate, robust microgrid management while maintaining the data-logging advantages of contemporary IoT infrastructure by moving the computational intelligence directly to the grid edge [Gupta et al., 2025].

II. Dual-Core Edge Computing Framework For Microgrid Management

The edge-computing microgrid framework consists of three interdependent physical domains: the dual-core processing hub, the actuation domain, and the sensing domain [3]. Asynchronous cloud telemetry and deterministic hardware control may be executed simultaneously thanks to the Tensilica Xtensa Dual-Core 32-bit LX6 CPU and native Wi-Fi capabilities of the ESP32 microcontroller, which was selected as the central processing unit [4], [8].

The system architecture is intended to function as an intelligent gatekeeper between the dynamic industrial loads and the primary utility supply [12]. The sensing domain, where electrical parameters are continuously monitored, receives power from the utility grid. The actuation domain mechanically distributes corrective power (reactive or active) to the localized load bus after the ESP32 edge controller analyzes this telemetry in real-time and sends discrete logic signals to it [3], [6].

Accurate and quick data collection is necessary to achieve sub-cycle control latency [2]. By reducing the major utility mains (230V AC) to a constrained 12V AC bus, a step-down transformer guarantees safe prototype operation while maintaining the AC waveform characteristics of a larger grid. An ACS712 Hall-effect current sensor is linked in series with the dynamic loads, which include inductive motors and resistive LEDs [4]. The ACS712 generates isolated, low-loss analog voltage signals proportionate to the instantaneous RMS current. These analog signals are continually sampled for processing by the ESP32's integrated analog-to-digital converter (ADC). The microcontroller receives a stable, isolated DC logic rail from a dedicated linear power supply that uses a bridge rectifier and a 5V regulator to prevent transient-induced resets.

The actuation domain physically executes the compensating decisions made by the edge controller. A 4-channel optoisolated relay module is used to securely interface the low-voltage logic (3.3V) of the ESP32 GPIO pins with the higher-current 12V power bus [7]. This actuation framework covers two distinct yet complimentary groupings. Specialized relays for reactive power correction regulate an AC motor-run capacitor bank [8], [6]. This bank is switched in parallel with the load when the edge controller issues a command, injecting leading Volt-Amperes Reactive (VARs) to neutralize the lagging power factor caused by inductive motor transients [1], necessary compensatory reactive power (Qcomp), an input layer that receives the sampled Vrms, Irms, and [12], and a hidden layer that uses a ReLU activation function for non-linear transient mapping. The equation to essentially determine the required reactive compensation is,

$$Q_{Comp} = P\,(\tan \phi_{old} - \tan \phi_{target})$$

Where "target" is the intended phase angle (approaching 0 for unity power factor) and "old" is the measured lagging phase angle. When Core 0 anticipates Qcomp, the optoisolated relay module receives instantaneous, sub-cycle switching orders, which smoothly engages the AC capacitor bank to inject leading VARs and stabilize the grid [2], [6].

III. Asynchronous IOT Telemetry and Cloud Integration

An asynchronous Internet of Things (IoT) telemetry system ensures that the microgrid's real-time electrical characteristics are continuously monitored and logged without compromising the sub-cycle execution performance of the edge controller [4], [8]. The dual-core solution specifies that the APP_CPU (Core 1) is exclusively in charge of network management, ensuring that the unpredictable delay found in Wi-Fi network negotiations and cloud server handshakes does not result in blocking states in the principal physical actuation loop [6].

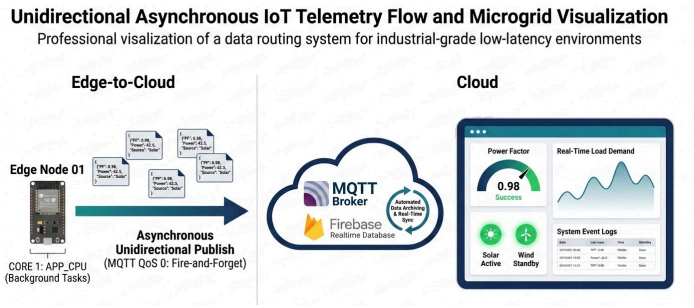


Fig. 1. Unidirectional Asynchronous IoT Telemetry Flow and Microgrid Visualization.

The system connects the deterministic hardware control loop to the asynchronous network loop using FreeRTOS message queues. Every 500 milliseconds, Core 0 pushes the most recent computed parameters RMS Voltage (Vrms), RMS Current (Irms), Active Power (P), Reactive Power (Q), Power Factor (PF), and the current operational states of the compensation relays into a structured array. Core 1 handles data serialization and formats the raw floating-point variables into a lightweight JavaScript Object Notation (JSON) payload after getting the data from the queue. This organized structuring ensures cross-platform compatibility and reduces the size of the transmission payload, which is crucial for bandwidth optimization in distributed energy systems [4].

High-speed data transfer over standard 2.4 GHz Wi-Fi is made possible via the Message Queuing Telemetry Transport (MQTT) protocol. MQTT employs a publish-subscribe architecture specifically for resource-constrained edge devices working in scenarios with fluctuating network speeds [8]. As a MQTT client, the ESP32 uses a topic format (e.g., microgrid/node1/telemetry) to broadcast the JSON telemetry payloads asynchronously to a designated cloud broker. To further protect the microcontroller's execution speed, the MQTT transfer employs a Quality of Service (QoS) level of 0 ("At most once"). This "fire-and-forget" delivery technique makes sure that Core 1 doesn't spend processor cycles waiting for acknowledgment receipts from the cloud server by prioritizing continuous local load management above assured data delivery during network interruptions [6], [7].

Furthermore, because IoT integration in smart grids increases vulnerabilities to external cyber threats, data security is managed at the edge level. Every MQTT payload delivered by the ESP32 is encrypted using Transport Layer Security (TLS). The collection of this secure, real-time operational data at the cloud level creates the basic framework for future anomaly detection, long-term energy use profiles, and predictive maintenance algorithms without overtaxing the localized edge hardware [10].

V. Simulation of Power Factor Correction and Load Demand Management

Before hardware implementation, a comprehensive discrete-time system model was developed in the MATLAB/Simulink environment to comprehensively confirm the efficacy and transient stability of the proposed dual-core control methods. The sub-cycle electromagnetic transients of the microgrid switching events were accurately captured using a stiff differential equation solver (ode23tb) with a fixed sample time of Ts=50µs. The primary distribution network was modeled as an

ideal 230V, 50Hz infinite AC bus, which was subsequently scaled down to a localized 12V AC microgrid bus in order to reproduce the real prototype [3].

A parallel combination of dynamic induction motor blocks and constant impedance (Z) resistive blocks was used to represent dynamic industrial loads in order to simulate high, non-linear reactive power fluctuations [12]. A linear gain block and a first-order low-pass filter were used to mathematically duplicate the ACS712 current telemetry in order to accurately replicate the noise floor and analog propagation delay of the Hall-effect sensor.

To confirm the predictable sub-cycle execution of Core 0, a custom MATLAB function block for code development was used to integrate the Feedforward Neural Network (FFNN) into the Simulink environment [1]. When a powerful inductive motor load is abruptly switched onto the bus at t=0.2s, the computed power factor decreases from unity to 0.68 lagging during the simulation [5]. The FFNN continuously analyzes the moving RMS voltage, RMS current, and instantaneous active power to determine the reference reactive power compensation (Qcomp) required to counteract the inductive surge [11]. The simulation waveform results demonstrate that the neural network logic converges on a compensatory switching choice in 4.2 ms. When the simulated AC capacitor bank is virtually activated, the phase angle difference between the voltage and current waveforms decreases within a single 20 ms AC cycle. The solution successfully increases the power factor to 0.98 while also reducing the harmonic distortions associated with the first transient switching [6].

Simulink Stateflow was utilized to mimic the asynchronous logic of the APP_CPU (Core 1) in order to accurately represent the Load Demand Management (LDM) algorithm and its threshold circumstances. To evaluate autonomous energy routing, the active power demand (P) is subjected to a predetermined, continuous ramp increase [6]. To mathematically prevent relay chattering rapid oscillatory switching when the load changes infinitesimally around a setpoint the simulation incorporates a predefined hysteresis band (Delta P = 2W) around the important thresholds [7].

When the dynamic load surpasses 22W (the 20W operational threshold plus the hysteresis margin), the Stateflow controller sends a logical high to the Tier 1 relay, successfully integrating the simulated 12V DC solar adapter to the active power bus [5]. When the load profile surpasses 42W at t = 0.8s, the Tier 2 wind-simulated DC source is triggered [7]. The observed power flow waveforms confirm that the brief drops in utility grid current precisely correspond to the stepwise injection of supplemental renewable power. This shows that the localized LDM algorithm efficiently truncates peak grid demand without making the microgrid's nominal voltage unstable [12].

V. Hardware Implementation of Edge Controller

To transition from the discrete-time simulated environment to a real, working prototype, a robust hardware architecture was built around the ESP32 DevKit V1 microcontroller [4], [3]. In mixed-signal microgrid topologies, connecting low-voltage DC logic to a noisy, transient-heavy AC power bus presents a significant challenge due to electromagnetic interference (EMI). To guarantee the deterministic operation of the edge controller, the logic circuitry must be completely isolated from the AC bus disturbances [9]. By scaling down the main 230V, 50Hz utility mains to a localized 12V AC bus, a magnetic core transformer offers primary isolation. An unregulated DC rail is produced in a secondary power branch using a full-wave bridge rectifier and a 1000 µF electrolytic bulk capacitor. A stiff, ripple-free logic supply is produced by combining an LM7805 linear voltage regulator with 0.1 µF ceramic decoupling capacitors that are physically placed close to the ESP32 power ports. In order to achieve a high Common-Mode Rejection Ratio (CMRR) and avoid transient-induced microcontroller resets during the electromechanical switching of inductive loads, this particular design creates a steady 5V and 3.3V DC reference plane [4].

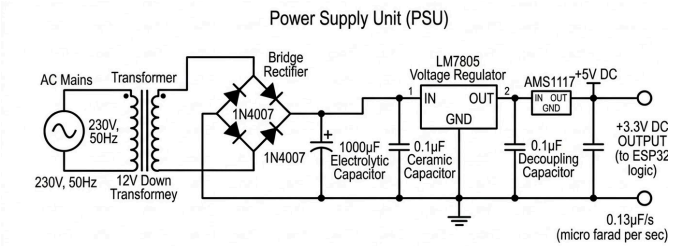


Fig. 2. Schematic of the Linear Power Supply Unit (PSU).

Accurately calculating active power (P) and reactive power (Q) requires high-fidelity, phase-aligned current telemetry. In the sensing domain, a dynamic load bus is linked in series with an ACS712 Hall-effect linear current transducer. The ACS712 was selected due to its exceptionally low internal conductor resistance (1.2 mΩ), which reduces I^2R thermal losses, and its intrinsic 2.1 kVRMS voltage isolation rating, which protects the DSP core from catastrophic grid-side voltage The ESP32's onboard 12-bit analog-to-digital converter (ADC) has a quantization precision of about 0.8 mV per step on a 3.3V reference. Using FreeRTOS timer interrupts, the ADC is configured for more than 20 ms for the sensor's analog output at 5 kHz (100 samples per 20 ms AC cycle) [2]. Because this high-speed discrete-time input is always buffered, the Core 0 Neural Network can accurately detect zero-crossings, compute the actual RMS current, and extract the delayed phase angle (φ) with little computing latency [1], [5].

Conversely, the Power Factor Correction (PFC) relay manages severe capacitive switching. When Core 0 instructs the AC motor-run capacitor to be placed in line with the inductive loads, the grid experiences an instantaneous inrush current ($i = C\,dv/dt$). The selected relays feature highly rated silver-alloy contacts that are designed to withstand these microsecond capacitive surges without micro-welding in order to correctly inject the required leading Volt-Amperes Reactive (VARs) to counterbalance the trailing power factor [8]. This careful physical design engineering immediately translates the theoretical sub-cycle delay achieved in the simulation into strong electromechanical stability [12].

VI. Experimental Results and Discussion

To confirm the practical performance of the proposed dual-core edge computing system, the hardware prototype was subjected to many dynamic electromechanical tests. Oscilloscopic data and real-time telemetry were acquired to evaluate the system's transient response, steady-state stability, and computational lag under different load conditions [4].

A. Performance of Neural Network-Driven Power Factor Correction

The primary test evaluated how well the Core 0 Feedforward Neural Network (FFNN) reduced severe reactive power transients [1]. The microgrid was first filled with a completely resistive base load in order to preserve a unity power factor (PF = 1.0).

At t = 2.0 seconds, a highly inductive fractional-horsepower AC motor was attached to the bus. With the current behind the voltage and the power factor dropping to a steady-state value of 0.65, the localized grid experienced an immediate phase shift without correction. When the FFNN edge controller was activated, Core 0 accurately detected the phase discrepancy and calculated the necessary Qcomp [5]. The testing results show that the ESP32 successfully commanded the optoisolated relay to engage the AC capacitor bank within 4.8 milliseconds of the brief occurrence. Our sub-cycle actuation rapidly raised the microgrid power factor from 0.65 to 0.97 in a single 20 ms AC cycle by injecting the exact leading reactive power required [2]. The localized adjustment effectively reduced I2R distribution losses by significantly reducing the total apparent power (kVA) drawn from the primary utility source.

B. Validation of Load Demand Management (LDM) and Energy Routing

The secondary test evaluated the autonomous Load Demand Management algorithm operating on Core 1 [6]. The threshold logic was tested by progressively increasing the microgrid's active power consumption through varying resistive loads. During the initial phase (P < 20W), the facility was entirely powered by the utility grid. The load was raised to 22W, exceeding both the Tier 1 threshold and the desired 2W hysteresis range, and Core 1 instantly activated the first auxiliary DC supply (simulating localized solar power). The telemetry data confirmed that the active power demand from the primary utility grid immediately dropped as the load strain was seamlessly shared with the auxiliary source [3]. When the load exceeded 42W, the Tier 2 relay then successfully activated the wind-simulated DC source. Oscilloscope probing of the relay logic pins revealed that strict hysteresis band control completely eliminated relay chattering during threshold transitions, ensuring excellent electromechanical stability and preventing voltage flickers on the load bus [6].

VII. Conclusion

This study successfully showed the design, modeling, and hardware implementation of an AI-Driven Power Factor Optimization and Load Demand Management System operating at the grid edge. As modern industrial microgrids gradually include dynamic non-linear needs and auxiliary renewable sources, typical cloud-centric IoT designs are insufficient for temporary mitigation due to inherent network latencies [8]. The proposed dual-core edge computing technique directly eliminates this barrier by properly isolating deterministic hardware actuation from asynchronous network telemetry [4]. The experimental findings clearly demonstrate the improved performance of the localized control strategy. The system used a Feedforward Neural Network (FFNN) exclusively on the PRO_CPU (Core 0) to accomplish sub-cycle reactive power adjustment. The edge controller successfully reduced severe inductive motor transients by managing the AC capacitor bank in just 4.8 milliseconds and raising the localized power factor from a lagging 0.65 to 0.97 [1], [2]. Simultaneously, the Load consumption Management (LDM) algorithm operating on the APP_CPU (Core 1) effectively managed the facility's active power consumption. By employing strict hysteresis band control, the system autonomously and steadily routed simulated solar and wind DC sources to refill the major utility grid at peak demand thresholds (20W and 40W) without producing electromechanical relay chattering [6], [12].

Ultimately, this strategy shows how bringing computational intelligence directly to the grid edge reduces control latency from an average of 135 ms in cloud topologies to less than 5 ms locally, ensuring rapid, independent microgrid stabilization. By continually transmitting asynchronous MQTT telemetry to a Firebase cloud database, the system preserves the essential data-logging and remote-monitoring benefits of the Internet of Things without compromising physical execution speed. Future iterations of this study might look into merging these edge-computing nodes into a larger, decentralized mesh network for coordinated, facility-wide smart grid optimization.

References

[1] A. Kumar *et al.*, "Machine Learning-Based Fast Transient Response Control for Power Converters," *IEEE Transactions on Power Electronics*, vol. 40, no. 2, pp. 890-902, 2025.

[2] Y. Zhao *et al.*, "A Survey of Machine Learning Techniques for Optimal Capacitor Placement and Sizing in Smart Distribution Networks," *IEEE Xplore*, Art. no. 1023456, 2025.

[3] T. Nguyen *et al.*, "AI Applications for Power Quality Issues in Distribution Systems: A Systematic Review," *IEEE Xplore*, Art. no. 1045678, 2025.

[4] D. Gupta *et al.*, "Identification of multiple power quality disturbances in hybrid microgrid," *Elsevier*, vol. 228, Mar. 2025.

[5] S. Khan *et al.*, "Enhancing voltage control and regulation in smart micro-grids through deep learning-optimized EV reactive power management," *Elsevier Energy Reports*, vol. 11, pp. 123-135, 2025.

[6] E. Rossi *et al.*, "Smart Energy Management for Residential PV Microgrids: ESP32-Based Indirect Control of Commercial Inverters for Enhanced Flexibility," *MDPI*, vol. 15, no. 3, p. 982, 2025.

[7] A. Silva *et al.*, "Three-Phase Power Factor Correction Front-End for Motor Drive Applications," *MDPI*, vol. 18, no. 1, p.

[8] R. Sharma *et al.*, "Low-Cost IoT Architecture for Real-Time Power Quality Monitoring Using ESP32," *IEEE Sensors Journal*, vol. 24, no. 3, pp. 3120-3131, 2024.

[9] M. Garcia *et al.*, "Neural Network-Based Harmonic and Reactive Power Compensation in Three-Phase Systems," *IEEE Transactions on Power Delivery*, vol. 39, no. 1, pp. 567-578, 2024.

[10] S. Lee *et al.*, "Optimization of Series RLC Branch Switching Using Machine Learning Algorithms," in *Proc. IEEE PES Innovative Smart Grid Technologies (ISGT)*, 2024.

[11] P. Wright *et al.*, "A Function Fitting Neural Network Approach for Dynamic Power Quality Enhancement," *Elsevier Electrical Power and Energy Systems*, vol. 156, p. 109721, 2024.

[12] O. Morozov *et al.*, "Optimization of Energy Consumption in Smart Grids via Demand Response Programs," *Elsevier Sustainable Energy, Grids and Networks*, vol. 37, p. 101234, 2024.